

Control vs Type 2 Diabetes Single-Cell Transcriptomics

Datasets, literature, learning roadmap and analysis plan

Compiled 2026-10-05. Every item carries a verification label: VERIFIED (read in paper text), ABSTRACT (abstract only), UNVERIFIED (from memory; check before use). GEO, NCBI and EBI pages could not be opened from the compiling environment, so no GEO sample metadata has been confirmed.

Contents: 1 Links; 2 Dataset verification log; 3 Reading list part 1; 4 Reading list part 2 (gaps); 5 Mentor plan part 1 (datasets, papers, 12-week roadmap); 6 Mentor plan part 2 (workflow, aims, portfolio, mistakes, glossary).

Links: datasets and priority papers

Links built from accessions/DOIs/PMIDs returned by searches or read in paper text. I could not open GEO, ArrayExpress or PANC-DB from my sandbox, so the dataset pages themselves are unchecked; click through and confirm sample labels.

Datasets

Dataset	Accession	Link	Status
Lawlor 2017 (5 ND, 3 T2D)	GEO GSE86473	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE86473	Accession read in paper
Lawlor 2017 raw	SRA SRP075970, BioProject PRJNA323853	https://www.ncbi.nlm.nih.gov/bioproject/PRJNA323853	Accession read in paper
Avrahami 2020 (10 T2D adults)	GEO GSE154126	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE154126	Accession read in paper
Segerstolpe 2016 (6 healthy, 4 T2D)	Web portal	http://sandberg.cmb.ki.se/pancreas	URL read in paper
Segerstolpe 2016	ArrayExpress E-MTAB-5061	https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-5061	UNVERIFIED accession (from memory)
Xin 2016	GEO GSE81608	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE81608	UNVERIFIED accession (from memory)
HPAP (Elgamal 2023)	PANC-DB	https://hpap.pmacs.upenn.edu	UNVERIFIED URL (from memory); paper says raw FASTQ come from the PANC-DB portal
Bandesh 2025/2026	see paper's data availability	https://pmc.ncbi.nlm.nih.gov/articles/PMC11785113/	Accession not retrieved

Papers (reading order)

#	Paper	DOI link	PMC (free full text)
1	Luecken & Theis, Current best practices in scRNA-seq analysis (Mol Syst Biol 2019)	https://doi.org/10.15252/msb.20188746	https://pmc.ncbi.nlm.nih.gov/articles/PMC6582955/
2	Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	https://doi.org/10.1038/s41576-023-00586-w	https://pmc.ncbi.nlm.nih.gov/articles/PMC10066026/
3	Confronting false discoveries in single-cell DE (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-25960-2	https://pmc.ncbi.nlm.nih.gov/articles/PMC8479118/
4	Benchmarking differential state methods, multi-subject scRNA-seq (Brief Bioinform 2022)	https://doi.org/10.1093/bib/bbac286	https://pmc.ncbi.nlm.nih.gov/articles/PMC9487674/
5	Design and power analysis for multi-sample single cell genomics (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-26779-7	https://pmc.ncbi.nlm.nih.gov/articles/PMC8595682/
6	Elgamal et al., Integrated map of islet gene expression (Diabetes 2023)	https://doi.org/10.2337/db23-0130	https://pmc.ncbi.nlm.nih.gov/articles/PMC10588282/
7	Lawlor et al. (Genome Res 2017)	https://doi.org/10.1101/gr.212720.116	https://pmc.ncbi.nlm.nih.gov/articles/PMC5287227/
8	Segerstolpe et al. (Cell Metab 2016)	https://doi.org/10.1016/j.cmet.2016.08.020	https://pmc.ncbi.nlm.nih.gov/articles/PMC5069352/
9a	Benchmarking atlas-level data integration (Nat Methods 2022)	https://doi.org/10.1038/s41592-021-01336-8	https://pmc.ncbi.nlm.nih.gov/articles/PMC8748196/
9b	Batch correction methods are often poorly calibrated (Genome Res 2025)	https://doi.org/10.1101/gr.279886.124	https://pmc.ncbi.nlm.nih.gov/articles/PMC12315870/
10a	Milo (Nat Biotechnol 2022)	https://doi.org/10.1038/s41587-021-01033-z	https://pmc.ncbi.nlm.nih.gov/articles/PMC7617075/
10b	scCODA (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-27150-6	https://pmc.ncbi.nlm.nih.gov/articles/PMC8616929/

Extra, as you reach those weeks

Paper	DOI link	PMC
Avrahami et al. 2020	https://doi.org/10.1016/j.molmet.2020.101057	https://pmc.ncbi.nlm.nih.gov/articles/PMC7471622/
Xin et al. 2016	https://doi.org/10.1016/j.cmet.2016.08.018	none returned

Paper	DOI link	PMC
Bandesh et al. (bioRxiv 2025)	https://doi.org/10.1101/2025.01.17.633590	https://pmc.ncbi.nlm.nih.gov/articles/PMC11785113/
Bandesh et al. (EMBO J 2026)	https://doi.org/10.1038/s44318-026-00744-w	https://pmc.ncbi.nlm.nih.gov/articles/PMC13226668/
Islet cell identity genesets (Nat Commun 2022)	https://doi.org/10.1038/s41467-022-29588-8	https://pmc.ncbi.nlm.nih.gov/articles/PMC9019032/
decoupleR (Bioinform Adv 2022)	https://doi.org/10.1093/bioadv/vbac016	https://pmc.ncbi.nlm.nih.gov/articles/PMC9710656/
Integrated single cell and spatial map of human white adipose tissue (Nat Commun 2023)	https://doi.org/10.1038/s41467-023-36983-2	https://pmc.ncbi.nlm.nih.gov/articles/PMC10017705/

Control vs T2D single-cell datasets: verification log (work in progress)

Status as of 2026-10-05. Only facts read from a primary source (paper full text on PMC) are marked VERIFIED. GEO, NCBI, and EBI were unreachable from the sandbox, so GEO pages were NOT read directly. Anything marked UNVERIFIED comes from memory or abstracts and must be checked on GEO/ArrayExpress before use.

Pancreatic islets (human, scRNA-seq)

Dataset	Paper	Donors	Cells	Tech	Accession	Status
Lawlor 2017	Genome Res, doi:10.1101/gr.212720.116	8 (5 ND, 3 T2D)	1,050 (622 ND, 428 T2D)	Fluidigm C1, ~3M reads/cell	Raw: SRA SRP075970 / PRJNA323853. Processed: GEO GSE86473	VERIFIED (paper text)
Segerstolpe 2016	Cell Metab, doi:10.1016/j.cmet.2016.08.020	10 (6 healthy, 4 T2D)	2,209	Smart-seq2, ~750k reads/cell	Portal: sandberg.cmb.ki.se/pancreas. ArrayExpress E-MTAB-5061 UNVERIFIED	Counts VERIFIED; accession UNVERIFIED
Avrahami 2020	Mol Metab, doi:10.1016/j.molmet.2020.101057	22 (1 newborn, 5 toddler, 2 adolescent, 4 ND adult, 10 T2D adult)	not retrieved	Fluidigm C1, SMART-seq, median 1.7M aligned reads	GEO GSE154126	VERIFIED except cell count
Elgamal 2023 (HPAP integrated map)	Diabetes, doi:10.2337/db23-0130	67 downloaded (29 ND, 17 T2D, 10 T1D, 9 ND Aab+); 65 in final map	192,203	HPAP scRNA-seq; raw FASTQ from PANC-DB	PANC-DB portal (data access terms apply)	VERIFIED (paper text); 10x vs other chemistry not confirmed
Xin 2016	Cell Metab, doi:10.1016/j.cmet.2016.08.018	ND and T2D organ donors (counts not retrieved)	1,492 (alpha, beta, delta, PP)	not confirmed	GEO GSE81608 UNVERIFIED	Abstract only
Bandesh 2025/2026	bioRxiv doi:10.1101/2025.01.17.633590; EMBO J doi:10.1038/s44318-026-00744-w	not retrieved	not retrieved	not retrieved	not retrieved	Title/abstract only; read the paper

Notes for a beginner:

- Lawlor, Segerstolpe, Xin are small (4-10 donors, plate-based). Good for learning and for quick replication, weak for donor-level statistics.
- Elgamal/HPAP has the most T2D donors (17 vs 29 ND) and is the best candidate for the main analysis, but needs PANC-DB access and re-processing from FASTQ.

Adipose tissue

No human adipose scRNA/snRNA dataset with explicit T2D vs control labels has been verified yet. Leads (all UNVERIFIED for T2D labels, donor counts, and accessions):

- Single-cell analysis of human adipose tissue, depot and disease specific cell types, Nat Metab 2020, doi:10.1038/s42255-019-0152-6.
- Human adipose snRNA-seq study using dataset HRA002549 (GSA, China), obesity vs obesity+T2D macrophage analysis, doi:10.3760/cma.j.cn112137-20250324-00714.
- Insulin-resistance snRNA-seq in subcutaneous adipose, PLoS Genet 2020, doi:10.1371/journal.pgen.1009018.
- Emont et al. 2022 (Nature) human/mouse adipose snRNA-seq: obesity-focused, not T2D.

Benchmarks / methods leads (for part 2, not yet read)

- Benchmarking differential state methods for multi-subject scRNA-seq: Nat Commun/Brief Bioinform, doi:10.1093/bib/bbac286 (pseudobulk performed best).
- Pseudobulk with offsets vs GLMM: Bioinformatics, doi:10.1093/bioinformatics/btae498.

Next steps

1. Confirm GSE/E-MTAB accessions and per-sample labels on GEO/ArrayExpress (needs network access to those hosts).
2. Read Bandesh 2025/2026 and Elgamal methods for chemistry, QC, and donor metadata.
3. Search Single Cell Portal, HCA, and CELLxGENE for T2D-labelled adipose/liver/muscle data.

Reading list: Control vs T2D single-cell transcriptomics

Every entry below was returned by a literature index search (title, DOI, abstract) on 2026-10-05. I did NOT read full texts except where noted in datasets_verified_so_far.md. Takeaways are quoted or paraphrased from abstracts only. Verify details before citing. "Order" is a suggested reading order.

A. Foundational: workflow, QC, normalization

Order	Paper	DOI	Why read it
1	Current best practices in single-cell RNA-seq analysis: a tutorial (Mol Syst Biol 2019)	10.15252/msb.20188746	Start here. End-to-end workflow and the decisions at each step.
2	Tutorial: guidelines for the computational analysis of single-cell RNA sequencing data (Nat Protoc 2021)	10.1038/s41596-020-00409-w	Second pass with a protocol focus.
3	Bioconductor workflow for single-cell RNA sequencing: normalization, dimensionality reduction, clustering, lineage inference (F1000Res 2017)	10.12688/f1000research.12122.1	R/Bioconductor route if you choose R over Python.
4	Analysis and visualization of single-cell sequencing data with Scanpy and MetaCell: a tutorial (Methods Mol Biol 2024)	10.1007/978-1-0716-3642-8_17	Python/Scanpy route, matches your background.

B. Quality control: ambient RNA and doublets

Paper	DOI	Why read it
Decontamination of ambient RNA in single-cell RNA-seq with DecontX (Genome Biol 2020)	10.1186/s13059-020-1950-6	Ambient RNA method.
Understanding and mitigating the impact of ambient mRNA contamination (PLoS One 2025)	10.1371/journal.pone.0332440	Abstract states ambient transcripts appear among DEGs; compares CellBender and SoupX. Directly relevant to islets (insulin/glucagon spillover).
FastCAR: fast correction for ambient RNA to facilitate differential expression (BMC Genomics 2023)	10.1186/s12864-023-09822-3	Abstract: ambient RNA levels are highly sample-specific, which matters for case vs control.
Souporcell: clustering by genotype without reference genotypes (Nat Methods 2020)	10.1038/s41592-020-0820-1	Donor demultiplexing and cross-genotype doublets.

C. Integration and batch correction (needed because donors are batches)

Paper	DOI	Takeaway from abstract
A benchmark of batch-effect correction methods for scRNA-seq (Genome Biol 2020)	10.1186/s13059-019-1850-9	Recommends Harmony, LIGER, Seurat 3.
Benchmarking atlas-level data integration in single-cell genomics (Nat Methods 2022)	10.1038/s41592-021-01336-8	BBKNN, Scanorama, scVI do well on complex tasks.
Batch correction methods ... are often poorly calibrated (Genome Res 2025)	10.1101/gr.279886.124	Recommends only Harmony. Read next to the two above; they disagree.
Probabilistic harmonization and annotation with deep generative models, scVI/scANVI (Mol Syst Biol 2021)	10.15252/msb.20209620	scVI/scANVI model reference.

D. Case-control statistics (the most important section for your design)

Paper	DOI	Takeaway from abstract
Confronting false discoveries in single-cell differential expression (Nat Commun 2021)	10.1038/s41467-021-25960-2	Why cell-level tests inflate false positives. Read first.
Differential gene expression analysis for multi-subject scRNA-seq with aggregateBioVar (Bioinformatics 2021)	10.1093/bioinformatics/btab337	Naive testing gives many false discoveries; pseudobulk has better FDR control.
A balanced measure shows superior performance of pseudobulk methods over mixed models and pseudoreplication (Nat Commun 2022)	10.1038/s41467-022-35519-4	Supports pseudobulk.

Paper	DOI	Takeaway from abstract
Benchmarking methods for detecting differential states between conditions from multi-subject scRNA-seq (Brief Bioinform 2022)	10.1093/bib/bbac286	Abstract: pseudo-bulk methods performed generally best.
Pseudobulk with proper offsets has the same statistical properties as GLMMs (Bioinformatics 2024)	10.1093/bioinformatics/btae498	Counterpoint on mixed models.
Valid post-clustering differential analysis for scRNA-seq (Cell Syst 2019)	10.1016/j.cels.2019.07.012	Double-dipping after clustering.

E. T2D single-cell papers (islets)

Paper	DOI	Role
Xin et al., RNA sequencing of single human islet cells reveals T2D genes (Cell Metab 2016)	10.1016/j.cmet.2016.08.018	1,492 cells, ND vs T2D donors (abstract).
Segerstolpe et al., Single-cell transcriptome profiling of human islets in health and T2D (Cell Metab 2016)	10.1016/j.cmet.2016.08.020	6 healthy, 4 T2D, 2,209 cells (read in full).
Lawlor et al., Single-cell transcriptomes identify human islet cell signatures ... in T2D (Genome Res 2017)	10.1101/gr.212720.116	5 ND, 3 T2D, 1,050 cells (read in full).
Avrahami et al., Single-cell transcriptomics of human islet ontogeny ... beta-cell dedifferentiation in T2D (Mol Metab 2020)	10.1016/j.molmet.2020.101057	22 donors incl. 10 T2D (read in full).
Camunas-Soler et al., Pancreas patch-seq links physiologic dysfunction in diabetes to transcriptomic phenotypes (Cell Metab 2020)	10.1016/j.cmet.2020.04.005	Links function to expression; advanced.
Elgamal et al., An integrated map of cell type-specific gene expression in pancreatic islets (Diabetes 2023)	10.2337/db23-0130	192,203 cells, 17 T2D vs 29 ND (read in full). Best model for your own pipeline.
Bandesh et al., Single-cell decoding of human islet cell type-specific alterations in T2D (bioRxiv 2025; EMBO J 2026)	10.1101/2025.01.17.633590 ; 10.1038/s44318-026-00744-w	Recent, large; read methods.
Single-cell mRNA-regulation analysis reveals cell type-specific mechanisms of T2D (Nat Commun 2025)	10.1038/s41467-025-65060-z	Re-analysis of public islet scRNA-seq.
Human pancreatic alpha-cell heterogeneity ... SMOC1 as a beta-cell dedifferentiation gene (Nat Commun 2025)	10.1038/s41467-025-62670-5	Combines scRNA-seq and snRNA-seq.
Meta-analysis of single-cell expression of human alpha and beta cells in T2D (J Diabetes 2021)	10.1111/1753-0407.13236	Example of combining the small public datasets.

F. Adipose tissue (relevant to this repo)

Paper	DOI	Note
Single-cell analysis of human adipose tissue identifies depot and disease specific cell types (Nat Metab 2020)	10.1038/s42255-019-0152-6	scRNA-seq of SVF from obese individuals.
An integrated single cell and spatial transcriptomic map of human white adipose tissue (Nat Commun 2023)	10.1038/s41467-023-36983-2	Integrates ten studies; >60 subpopulations; a reference for annotation.
Human subcutaneous and visceral adipocyte atlases (Nat Genet 2025)	10.1038/s41588-024-02048-3	snRNA-seq depot atlases.
Spatial mapping reveals human adipocyte subpopulations with distinct sensitivities to insulin (Cell Metab 2021)	10.1016/j.cmet.2021.07.018	Insulin sensitivity link.
Integrative single-cell analysis of metabolic syndrome in adipose tissue (Diabetol Metab Syndr 2025)	10.1186/s13098-025-01975-3	Metabolic-syndrome snRNA-seq reanalysis.
Robust snRNA-seq reveals depot-specific dynamics in adipose remodeling during obesity (eLife 2025)	10.7554/eLife.97981	Mouse-focused nuclei isolation method; check species before use.
Causal effect of obesity on prediabetes and insulin resistance (PLoS Genet 2020)	10.1371/journal.pgen.1009018	Bulk plus snRNA-seq, insulin resistance.

G. Atlases and portals

Paper	DOI
The Tabula Sapiens (Science 2022)	10.1126/science.abl4896
Tabula Sapiens 2.0 (bioRxiv 2024)	10.1101/2024.12.03.626516

Gaps (not yet found or not yet read)

- Experimental design and power for scRNA-seq case-control; sequencing technology comparisons (Smart-seq2 vs 10x vs snRNA-seq).
- Cell type annotation tools; cell-cell communication; pathway/GSEA on pseudobulk; trajectory inference.
- Your remaining items 3 to 5 (message was cut off).

Reading list, part 2: filling the gaps

Companion to literature_reading_list.md. Same caveat: every entry was returned by a literature-index search (title, DOI, abstract) on 2026-10-05; I did not read full texts. Takeaways come from abstracts only.

H. Experimental design and power

Paper	DOI	Why read it
Design and power analysis for multi-sample single cell genomics experiments (Nat Commun 2021)	10.1038/s41467-021-26779-7	Abstract: models sample size, cells per individual and depth. Core design paper.
Hierarchicell: power for differential expression with single-cell data (BMC Genomics 2021)	10.1186/s12864-021-07635-w	Abstract: existing calculators focus on total cells, not independent units (donors), and overestimate power.
Experimental design for single-cell RNA sequencing (Brief Funct Genomics 2018)	10.1093/bfpg/elx035	General design considerations.
Quantifying the tradeoff between sequencing depth and cell number in scRNA-seq (bioRxiv)	10.1101/762773	Budget allocation: cells vs depth.
Optimal design of scRNA-seq experiments for cell-type-specific eQTL analysis (Nat Commun 2020)	10.1038/s41467-020-19365-w	Abstract: different sample/cell/read designs can give similar power.
Robustness of scRNA-seq for identifying differentially expressed genes (BMC Genomics 2023)	10.1186/s12864-023-09487-y	Small cell numbers per cluster.

I. Technology comparison (matters for pooling Smart-seq2, C1, 10x, snRNA-seq datasets)

Paper	DOI	Takeaway from abstract
Direct comparative analysis of 10X Genomics Chromium and Smart-seq2 (Genomics Proteomics Bioinformatics 2021)	10.1016/j.gpb.2020.02.005	Smart-seq2 detected more genes.
Comparative analysis of single-cell RNA sequencing methods (Mol Cell 2017)	10.1016/j.molcel.2017.01.023	Smart-seq2 most sensitive per cell.
Full-length RNA-seq from single cells using Smart-seq2 (Nat Protoc 2014)	10.1038/nprot.2014.006	The protocol used by Segerstolpe.
Benchmarking single-cell mRNA-seq technologies ... cell types with low RNA content (2022)	10.7171/3fc1f5fe.dbeabb2a	Sensitivity in low-RNA cell types.

Gap: I did not find a dedicated single-nucleus vs single-cell comparison for adipose or islets in this search.

J. Cell type annotation

Paper	DOI	Why read it
Automated methods for cell type annotation on scRNA-seq data (Comput Struct Biotechnol J 2021)	10.1016/j.csbj.2021.01.015	Overview of approaches.
Evaluation of cell type annotation R packages on scRNA-seq data (Genomics Proteomics Bioinformatics 2021)	10.1016/j.gpb.2020.07.004	Benchmark of supervised tools.
Generation of human islet cell type-specific identity genesets (Nat Commun 2022)	10.1038/s41467-022-29588-8	Abstract: meta-analysis of scRNA-seq to define alpha/beta/gamma/delta marker sets. Use for islet annotation.
Transcriptomes of the major human pancreatic cell types (Diabetologia 2011)	10.1007/s00125-011-2283-5	Older bulk reference for alpha, beta, duct, acinar.
Integrated single cell and spatial map of human white adipose tissue (Nat Commun 2023)	10.1038/s41467-023-36983-2	Adipose annotation reference (also in section F).

Skipped: LLM-based annotation benchmark (Brief Bioinform 2025, 10.1093/bib/bbaf622). It exists, but I would not rely on it as a beginner.

K. Compositional / differential abundance (T2D may change cell-type proportions)

Paper	DOI	Takeaway from abstract
scCODA: a Bayesian model for compositional single-cell data analysis (Nat Commun 2021)	10.1038/s41467-021-27150-6	Handles compositionality and low sample size.
Milo: differential abundance testing using k-NN graphs (Nat Biotechnol 2022)	10.1038/s41587-021-01033-z	Neighbourhood-level DA without hard clusters.

Paper	DOI	Takeaway from abstract
Benchmarking differential abundance methods (Genome Biol 2024)	10.1186/s13059-023-03143-0	Compares DA methods.
Robust differential composition and variability analysis, sccomp (bioRxiv)	10.1101/2022.03.04.482758	Alternative composition model.
Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	10.1038/s41576-023-00586-w	Abstract: summarises benchmarks into best-practice workflows. Good umbrella paper to read alongside section A.

L. Pathways and regulator activity

Paper	DOI	Why read it
decoupleR: ensemble of methods to infer biological activities from omics data (Bioinform Adv 2022)	10.1093/bioadv/vbac016	One framework for pathway/TF activity; works on pseudobulk output.
PROGENy: perturbation-response genes reveal signaling footprints (Nat Commun 2018)	10.1038/s41467-017-02391-6	Pathway activity from footprint genes.
Transfer of regulatory knowledge from human to mouse (BBA Gene Regul Mech 2019)	10.1016/j.bbagr.2019.194431	DoRothEA and PROGENy benchmark.

Gap: search for single-cell-specific GSEA tools returned niche packages only. For T2D, running GSEA/decoupleR on pseudobulk DE results (per section D) is the safer route; I have no benchmark paper to cite for that choice.

M. Cell-cell communication

Paper	DOI	Takeaway from abstract
Inference and analysis of cell-cell communication using CellChat (Nat Commun 2021)	10.1038/s41467-021-21246-9	The CellChat method.
CellChat for systematic analysis of cell-cell communication ... (Nat Protoc 2024)	10.1038/s41596-024-01045-4	Protocol, good for a beginner.
Comparison of resources and methods to infer cell-cell communication from single-cell RNA data (Nat Commun 2022)	10.1038/s41467-022-30755-0	Compares resources and methods (LIANA). Read before choosing a tool.
ESICCC: evaluation, selection and integration of CCC inference methods (Genome Res 2023)	10.1101/gr.278001.123	Abstract: benchmarks 18 ligand-receptor methods.
Comparative analysis of cell-cell communication at single-cell resolution (Nat Biotechnol 2023)	10.1038/s41587-023-01782-z	Condition-comparison angle, useful for control vs T2D.

N. Trajectory / dedifferentiation

Paper	DOI	Why read it
A comparison of single-cell trajectory inference methods (Nat Biotechnol 2019)	10.1038/s41587-019-0071-9	Abstract: compares 29 methods.
Cell-state trajectories in T2D beta cells	see Avrahami 2020 and the SMOC1 alpha-cell paper in literature_reading_list.md section E	Applied examples of dedifferentiation analysis.

Index quirk: one search returned a record with this benchmark's abstract attached to an unrelated BMJ title (doi 10.1136/bmj.325.7378.1449). Ignore it; it is a metadata error in the index. Trajectory methods are optional for a first Control vs T2D analysis; treat as advanced.

O. Genetics link (T2D GWAS to cell types)

Paper	DOI	Why read it
Single cell chromatin accessibility reveals pancreatic islet cell type- and state-specific regulatory programs of diabetes risk (Nat Genet 2021)	10.1038/s41588-021-00823-0	Abstract: T2D GWAS enrichment in beta-cell states.
3D chromatin maps of the human pancreas reveal lineage-specific regulatory architecture of T2D risk (Cell Metab 2022)	10.1016/j.cmet.2022.08.014	Variant-to-gene links.
Cell-type-specific cis-eQTLs in pancreatic cell types identify novel risk genes for T2D (Brief Bioinform 2025)	10.1093/bib/bbaf531	Abstract: 328 cell-type-specific cis-eQTLs.
EPIC: inferring relevant cell types for complex traits from GWAS and scRNA-seq (PLoS Genet 2022)	10.1371/journal.pgen.1010251	Method to link GWAS to cell types.

Paper	DOI	Why read it
Implicating T2D effector genes with promoter-focused Capture-C (Diabetologia 2024)	10.1007/s00125-024-06261-x	Metabolic cell models.

Still open

- Single-nucleus vs single-cell comparisons specific to adipose or islets.
- A benchmark specifically supporting pathway analysis choice on single-cell/pseudobulk data.
- Your original items 3 to 5 (message was cut off before them).

Mentor plan, part 1: datasets, papers, 12-week roadmap, first analysis

Status 2026-10-05. Verification level is marked on every claim: VERIFIED = read in paper text; ABSTRACT = abstract only; UNVERIFIED = memory, must be checked. GEO/NCBI/EBI were unreachable from my sandbox, so no GEO page has been read directly.

1. Five datasets for Control vs T2D (human islets)

R a n k	Dataset	Donors (ND / T2D)	Cells	Tech	Access	Best for
1	Elgamal 2023, HPAP (doi:10.2337/db23-0130)	65 in final map; 29 ND, 17 T2D of 67 downloaded (also 10 T1D, 9 ND Aab+)	192,203	HPAP scRNA-seq (chemistry not confirmed)	Raw FASTQ via PANC-DB (data-access terms apply). Authors' processed object: UNVERIFIED	Main project
2	Lawlor 2017 (doi:10.1101/gr.212720.116)	8: 5 / 3	1,050 (622 ND, 428 T2D)	Fluidigm C1	SRA SRP075970; processed GEO GSE86473	Learning
3	Segerstolpe 2016 (doi:10.1016/j.cmet.2016.08.020)	10: 6 healthy / 4 T2D	2,209	Smart-seq2	Portal sandberg.cmb.ki.se/pancreas; E-MTAB-5061 UNVERIFIED	Validation, learning
4	Avrahami 2020 (doi:10.1016/j.molmet.2020.101057)	22 total: 4 adult ND, 10 adult T2D, 8 younger ND	not retrieved	Fluidigm C1	GEO GSE154126	Validation of beta-cell dedifferentiation
5	Bandesh 2025/2026 (doi:10.1101/2025.01.17.633590; 10.1038/s44318-026-00744-w)	not retrieved	not retrieved	not retrieved	not retrieved	Advanced; read methods first

Why not the others:

- Xin 2016 (1,492 cells, GSE81608 UNVERIFIED): donor counts not retrieved; small and older. Fine as extra validation once checked.
- No human adipose dataset with T2D vs control labels is verified. Adipose leads in literature_reading_list.md section F are obesity/insulin-resistance studies, not T2D case-control. Do not use adipose as the main tissue until a labelled dataset is confirmed.
- Donor reality check: Lawlor (3 T2D) and Segerstolpe (4 T2D) are too small for donor-level inference. Use them to learn the pipeline and for exploratory replication, not as evidence.
- Cell counts per donor are heavily uneven in these plate-based sets; check this before any proportion analysis.

2. Ten papers to read first (in order)

#	Paper	DOI	Read	Learn
1	Current best practices in scRNA-seq analysis: a tutorial (Mol Syst Biol 2019)	10.15252/msb.20188746	Whole	The full pipeline and its decisions.
2	Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	10.1038/s41576-023-00586-w	Skim, then reference	Benchmark-backed defaults.
3	Confronting false discoveries in single-cell differential expression (Nat Commun 2021)	10.1038/s41467-021-25960-2	Whole	Why cell-level tests fail; pseudobulk.
4	Benchmarking methods for differential states from multi-subject scRNA-seq (Brief Bioinform 2022)	10.1093/bib/bbac286	Whole	Abstract: pseudobulk best overall.
5	Design and power analysis for multi-sample single cell genomics (Nat Commun 2021)	10.1038/s41467-021-26779-7	Methods and figures	Donors, not cells, drive power.
6	Elgamal 2023 integrated islet map (Diabetes)	10.2337/db23-0130	Whole; copy the pipeline	A donor-aware T2D workflow.

#	Paper	DOI	Read	Learn
7	Lawlor 2017 (Genome Res)	10.1101/gr.212720.116	Results and methods	Your first dataset.
8	Segerstolpe 2016 (Cell Metab)	10.1016/j.cmet.2016.08.020	Results	Second dataset; compare findings.
9	Benchmarking atlas-level data integration (Nat Methods 2022)	10.1038/s41592-021-01336-8	Abstract, figures	When integration is needed. Read with the Genome Res 2025 critique (10.1101/gr.279886.124).
10	Milo (Nat Biotechnol 2022) and scCODA (Nat Commun 2021)	10.1038/s41587-021-01033-z ; 10.1038/s41467-021-27150-6	Intro and methods overview	Differential abundance.

Reviews from 2022-2026 on single-cell transcriptomics in diabetes: NOT yet found. I have not searched for reviews specifically; that is the next literature task.

3. Statistical stance (non-negotiable)

10 control + 10 T2D donors with 20,000 cells is $n = 20$, not 20,000. Cells from one donor share genotype, age, BMI, isolation batch, and ambient RNA, so they are nested within donors and correlated.

- Defensible for publication: pseudobulk (sum counts per donor per cell type) with edgeR, DESeq2 or limma-voom, with donor as the unit; or mixed models with a donor random effect, where the literature (see reading list section D) argues they perform comparably if specified well. Include covariates (sex, age, BMI, batch) when donor numbers allow.
- Not defensible: Wilcoxon or t-test across cells, reporting cell-level p-values as evidence of disease effect. Fine for marker finding within one dataset, not for Control vs T2D.
- Permutation tests: permute donor labels, never cell labels.
- Abundance: use donor-level proportions with a compositional method (scCODA, Milo, sccomp); proportions sum to 1, so one cell type rising makes others fall.

4. 12-week roadmap

Wk	Concepts	Read	Do	Output
1	DNA/RNA, transcription, exons/introns, RNA-seq, UMI, count matrix, MT/ribosomal genes	Paper 1 sections on experimental setup	Open a count matrix in pandas, compute library size, genes per cell	One-page glossary in your own words
2	Islet biology: alpha/beta/delta, insulin, glucose handling, T2D: beta-cell dysfunction, ER and oxidative stress, inflammation	Lawlor intro/discussion; Avrahami abstract	List marker genes per islet cell type (INS, GCG, SST, PPY, PRSS1, KRT19)	Marker table with sources
3	Cell vs nucleus, Smart-seq2 vs 10x, dropout, sparsity, depth, batch and donor effects	Paper 1; technology comparison papers (reading list part 2, section I)	Load Lawlor into AnnData; explore sparsity	Plots: counts/cell, genes/cell, MT fraction
4	QC, normalization, HVGs	Paper 1 QC/normalization sections	Run QC filters; justify thresholds per donor	Before/after QC figures, written thresholds
5	PCA, kNN graph, Leiden resolution, UMAP	Paper 1 clustering section	Cluster at several resolutions; check stability	Clustering notebook; resolution sweep
6	Marker genes, annotation	Islet identity genesets paper (10.1038/s41467-022-29588-8)	Annotate cell types; check for doublets and ambient hormone signal	Annotated UMAP, dotplot
7	Pseudoreplication, pseudobulk	Papers 3, 4	Pseudobulk by donor x cell type; edgeR/limma or pydeseq2; compare with a naive cell-level test	Volcano plots, inflation comparison
8	Power, abundance	Papers 5, 10	Donor-level proportions; scCODA or Milo	Abundance plot with donor points
9	Multiple testing, GSEA/GO/Reactome, decoupleR	decoupleR (10.1093/bioadv/vbac016)	Pathways on pseudobulk results	Enrichment figure, table
10	Replication	Seegerstolpe paper	Repeat steps on Seegerstolpe; compare direction of effects	Concordance plot, short report
11	Batch/integration, HPAP structure	Paper 9 and Genome Res critique	Request PANC-DB access; read Elgamal methods	Access request, data-plan note
12	Project design	Paper 6	Write aims and analysis plan	3-page proposal for your PI

5. Recommended first dataset and analysis

Dataset: Lawlor 2017 (GSE86473 processed; SRA SRP075970 raw). Reason: small, both labelled groups, plate-based so no droplet ambient-RNA complication, and the accession was read in the paper itself.

Weakness: only 3 T2D donors.

First analysis:

1. Load the processed matrix; confirm the donor and diagnosis labels from GEO sample metadata (I could not read them).
2. Per-donor QC, then normalize, HVGs, PCA, kNN, Leiden, UMAP.
3. Annotate alpha, beta, delta, PP, acinar, ductal; color UMAP by donor and diagnosis to see donor effects.
4. Pseudobulk by donor x cell type; test beta cells T2D vs ND with 5 vs 3 donors. Treat as exploratory.
5. Also run the naive cell-level test and compare; the gap is the lesson on pseudoreplication.

Tool choice: start with Scanpy/AnnData (you already know Python), and learn R only for the pseudobulk step if needed (edgeR/limma-voom or muscat are the field standard; I believe pydeseq2 is a Python alternative but I have not verified it this session). Learn Seurat later, mainly to read other people's code.

Open items

- Verify GEO metadata for GSE86473, GSE154126 and E-MTAB-5061, and the HPAP access path.
- Search for 2022-2026 reviews on single-cell transcriptomics in diabetes.
- Find a T2D-labelled adipose or other metabolic-tissue dataset.
- Remaining deliverables: top-10 dataset list, full workflow, aims, portfolio structure, mistakes list, 50-term glossary, step-by-step teaching notes.

Mentor plan, part 2: remaining deliverables

Verification labels: VERIFIED = read in paper text; ABSTRACT = abstract only; UNVERIFIED = memory, check before use. Deliverable 3 (12-week roadmap) is in mentor_plan_part1.md. Links are in links.md.

Deliverable 1: ranked public datasets for Control vs T2D

I cannot honestly give 10 verified datasets. I verified 4 from paper text and have leads for the rest. Do not treat leads as confirmed.

Rank	Dataset	Tissue	Donors (ND/T2D)	Cells	Status	Role
1	HPAP, Elgamal 2023, doi:10.2337/db23-0130	Islets	29 / 17 of 67 downloaded (65 in final map)	192,203	VERIFIED	Main
2	Lawlor 2017, GSE86473	Islets	5 / 3	1,050	VERIFIED	Learning
3	Segerstolpe 2016	Islets	6 / 4	2,209	VERIFIED counts; accession UNVERIFIED	Learning, replication
4	Avrahami 2020, GSE154126	Islets	4 adult ND + 4 younger groups / 10 T2D	not retrieved	VERIFIED donors	Replication
5	Bandesh 2025/2026, doi:10.1038/s44318-026-00744-w	Islets	not retrieved	not retrieved	ABSTRACT	Advanced
6	Xin 2016, GSE81608	Islets	not retrieved	1,492	ABSTRACT; accession UNVERIFIED	Extra replication
7	Camunas-Soler 2020 patch-seq, doi:10.1016/j.cmet.2020.04.005	Islets, with electrophysiology	not retrieved	not retrieved	ABSTRACT	Advanced (links function to expression)
8	Alpha-cell scRNA+snRNA-seq, doi:10.1038/s41467-025-62670-5	Islets	not retrieved	not retrieved	ABSTRACT	Advanced; snRNA-seq example
9	Sex differences, 52 donors with and without T2D, doi:10.21203/rs.3.rs-4607352/v1	Islets, scRNA + snATAC	52 total	not retrieved	ABSTRACT (preprint)	Validation if data are public
10	Adipose: no verified T2D-labelled set	Adipose	none	none	Gap	Leads in literature_reading_list.mdf

Unsuitable or caution:

- Datasets without diabetes labels (e.g. healthy-only islet atlases, mouse atlases) cannot answer your question.
- Mouse db/db or HFD islet data are not human T2D; use only for hypothesis support.
- Obesity-only adipose snRNA-seq studies do not isolate T2D; any adipose finding would be confounded by BMI.

Deliverable 2: 18 papers and resources (priority order)

Full DOIs and PMC links are in links.md. Short list: 1 Luecken & Theis best practices (whole). 2 Best practices across modalities (skim). 3 Confronting false discoveries (whole). 4 Multi-subject DS benchmark (whole). 5 Multi-sample power/design (methods, figures). 6 Elgamal 2023 (whole, copy the pipeline). 7 Lawlor 2017 (results, methods). 8 Segerstolpe 2016 (results). 9 Atlas-level integration benchmark + Genome Res 2025 critique (abstract, figures). 10 Milo + scCODA (methods overview). 11 Avrahami 2020 (dedifferentiation result). 12 Islet identity genesets (annotation markers). 13 decoupleR (tutorial). 14 CellChat Nat Protoc 2024 (protocol; optional). 15 Comparison of CCC methods (Nat Commun 2022; optional). 16 Bandesh 2025/2026 (methods, once you are on HPAP). 17 Camunas-Soler patch-seq (advanced). 18 Human adipose integrated map (adipose context). Reviews from 2022-2026 on single-cell transcriptomics in diabetes: NOT yet found; search PubMed for "single-cell diabetes review" before citing any.

Deliverable 4: end-to-end workflow

Teaching format for each step: what, why, assumptions, failure modes, validation, expected plot, how to say it to your PI.

4.1 Data and metadata

- What: one donor table (ID, diagnosis, sex, age, BMI, batch, isolation site, culture time) and one count matrix per dataset.
- Why: all inference is donor-level; metadata drives the model.
- Fail: batch perfectly aligned with diagnosis (then disease and batch cannot be separated).
- Validate: cross-tab diagnosis x batch x sex.
- Plot: donor-level bars of cell counts per donor.
- PI line: "Here are the donors; this is what we can and cannot adjust for."

4.2 QC

- What: remove empty droplets, dying cells, doublets; correct ambient RNA in droplet data.
- Why: low-quality cells form fake clusters; ambient INS/GCG makes alpha cells look INS-positive and creates false DEGs (ambient-RNA abstracts in `literature_reading_list.md` section B).
- Assumptions: thresholds are tissue-specific; do not copy PBMC cutoffs. Check MT fraction per donor and cell type.
- Fail: filtering with one global threshold that removes a whole donor or cell type.
- Validate: per-donor distributions before/after; cells removed per donor.
- Plot: violin of genes/UMIs/MT% per donor.
- PI line: "QC removed X% of cells, evenly across diagnosis."

4.3 Normalization

- What: put cells on a comparable scale. Standard: divide by total counts, scale, log1p. Alternatives: Pearson residuals; for Smart-seq2, length-normalized values.
- Why: cells differ in capture efficiency and depth, not biology.
- Assumption: most genes are unchanged and cell types have similar total RNA. Violated when a cell type has much more RNA.
- Fail: normalized values used for pseudobulk testing (use raw counts there).
- Validate: library size no longer drives PC1.
- PI line: "Normalization removes depth, not biology."

4.4 HVG, PCA, neighbours, clustering, UMAP

- HVG: keep genes with variance beyond expectation (about 2000), reducing noise.
- PCA: compress to 30-50 components; check an elbow plot and whether PCs track donor rather than biology.
- kNN graph: each cell links to its k nearest neighbours in PC space (try k 15-30).
- Leiden: finds densely connected groups on that graph. Resolution sets granularity: higher means more, smaller clusters.
- UMAP: 2D picture for display only. Distances and cluster sizes in UMAP are not quantitative; never test on UMAP coordinates.
- Fail: choosing resolution until you see a "T2D cluster" (circular).
- Validate: run several resolutions; check cluster stability, marker specificity, and that clusters contain multiple donors.
- Plot: UMAP coloured by cell type, donor, diagnosis side by side.
- PI line: "Clusters are hypotheses until markers and replication support them."

4.5 Annotation

- Use canonical markers (INS beta, GCG alpha, SST delta, PPY PP, PRSS1 acinar, KRT19 ductal, plus endothelial/stellate/immune) and the islet identity genesets paper; optionally reference mapping.
- Validate: dotplot; hormone ambient check; compare proportions with the original paper.

4.6 Cell-type abundance

- Unit: donor. Compute proportions per donor, then use scCODA, Milo (donor as replicate) or sccomp.

- Why: proportions sum to 1, so one type rising makes others fall.
- Fail: chi-square test on cell counts (pseudoreplication).
- Plot: per-donor proportions as points with group means.

4.7 Differential expression: Control vs T2D

- Default: pseudobulk. For each cell type, sum raw counts per donor; filter cell types with too few cells per donor and too few donors per group; model counts with edgeR (quasi-likelihood), DESeq2 or limma-voom; design: diagnosis + sex + age (+ batch/BMI if donors allow).
- Alternative: mixed model with donor random effect (e.g. NEBULA, glmmTMB, dreamlet). The Bioinformatics 2024 paper abstract says pseudobulk with proper offsets matches GLMMs; use mixed models mainly when donors contribute varying cell numbers and you want covariates handled per cell.
- Report: log2 fold change, raw p, FDR (Benjamini-Hochberg) per cell type, and decide how to handle multiplicity across cell types.
- Fail: p-values from per-cell Wilcoxon; treating UMAP separation as evidence; unadjusted covariates.
- Validate: p-value histogram is flat plus a spike near 0; MA/volcano plots; leave-one-donor-out stability; permutation of donor labels.
- Elgamal found 84 T2D genes across cell types with 17 T2D vs 29 ND (VERIFIED), so expect modest effects, not hundreds.
- PI line: "Effective sample size is donors; here is the donor-level result."

4.8 Pathways and regulators

- Run GSEA/GO/Reactome/KEGG on the ranked pseudobulk statistics (not on a thresholded list when possible); decoupleR for TF and pathway activity.
- Fail: interpreting enrichment from a ranked list built on cell-level p-values.

4.9 Replication

- Repeat the identical pipeline in an independent cohort; report direction concordance and effect-size correlation of DEGs, not just overlap counts.
- Caveat: platforms differ (Smart-seq2/C1 vs droplet), so compare signs and ranks, not magnitudes. Small cohorts have little power, so lack of replication is weak evidence.

4.10 Advanced, later

- Trajectory/pseudotime, RNA velocity, integration across datasets, scATAC, spatial. Not needed for the first project.

Deliverable 5: research question and aims

Question: How does type 2 diabetes alter cell-type composition and cell-type-specific transcription in human pancreatic islets, and are these changes reproducible across independent donor cohorts?

- Aim 1, cell-type identification: annotate islet cell types in HPAP using canonical markers; evaluate against published labels. Analysis: clustering stability, marker specificity.
- Aim 2, abundance: test donor-level proportion differences (scCODA/Milo/sccomp), adjusting for sex and age. Statistics: compositional model with donor as replicate; FDR across cell types.
- Aim 3, cell-specific DE: pseudobulk edgeR/limma-voom per cell type with covariates. Statistics: FDR per cell type; leave-one-donor-out sensitivity.
- Aim 4, pathways: GSEA/decoupleR on ranked pseudobulk statistics for beta, alpha, delta cells; check stress, inflammation, mitochondrial themes against literature.
- Aim 5, replication: repeat Aims 3-4 in Lawlor, Segerstolpe and Avrahami; test directional concordance (sign test, rank correlation).

Realism check: HPAP access and FASTQ processing may take weeks. A safe internship scope is Aims 1-3 on one dataset plus Aim 5 on a small one.

Deliverable 6: portfolio structure

t2d-islet-sc/	
README.md	question, data, key results, how to run
environment.yml	pinned versions (python, scanpy, edgeR via rpy2 or R env)

```

data/
  README.md          source URL, accession, download date, checksums
  raw/               never edited (git-ignored)
  metadata/donors.csv donor table
notebooks/
  01_qc.ipynb ... 08_replication.ipynb
src/                 reusable functions (QC, pseudobulk)
results/figures, results/tables
reports/             2-page summary, slide deck for PI
docs/                literature notes, decisions log (why each threshold)

```

Practices: fixed random seeds, one notebook per step, a decisions log, no large data in git, a short methods paragraph per analysis.

Deliverable 7: common mistakes (statistically invalid ones first)

Invalid:

1. Treating cells as independent replicates (pseudoreplication).
2. Using per-cell Wilcoxon p-values to claim Control vs T2D differences.
3. Permuting cell labels instead of donor labels.
4. Chi-square on cell counts for abundance.
5. Pseudobulk built from normalized/logged values instead of raw counts.
6. Batch or site perfectly confounded with diagnosis, and not stating it.
7. Clustering, then testing the clusters for diagnosis differences on the same data without acknowledging circularity (Cell Syst 2019 paper on post-clustering inference).
8. Integration that removes the disease signal, then claiming no effect.
9. Ignoring multiple testing across genes and cell types.
10. Reporting a cell type with 3 cells in a donor as a donor-level measurement.

Common technical:

11. Copying QC thresholds from other tissues.
12. Ignoring ambient hormone RNA in islets.
13. Reading UMAP distances and cluster sizes quantitatively.
14. Tuning resolution to get the cluster you want.
15. Annotating by one marker gene.
16. Ignoring sex, age, BMI and donor-level covariates.
17. Overinterpreting a p-value without effect size or replication.
18. Skipping donor-level plots.
19. Reporting pathway enrichment from an arbitrary thresholded list.
20. Mixing Smart-seq2 and droplet data without addressing technology effects.

Deliverable 8: glossary (50 terms)

1 Transcriptome: all RNA in a sample. 2 Gene expression: RNA produced from a gene. 3 Transcript: an RNA copy of a gene. 4 Exon: retained coding segment. 5 Intron: removed segment (reads from nuclei often include introns). 6 mRNA. 7 Poly(A): tail used for capture in most protocols. 8 Read: one sequenced fragment. 9 FASTQ: raw read file with quality. 10 Alignment: mapping reads to a genome. 11 UMI: tag that counts original molecules. 12 Cell barcode: tag identifying the cell. 13 Count matrix: genes x cells table of counts. 14 Library size: total counts in a cell. 15 Sequencing depth: reads per cell. 16 Dropout: gene present but not detected. 17 Sparsity: fraction of zeros. 18 scRNA-seq: whole cells. 19 snRNA-seq: nuclei; used for tissues like adipose. 20 Smart-seq2: full-length plate protocol. 21 10x Chromium: droplet protocol. 22 Droplet: oil bubble holding one cell. 23 Empty droplet: no cell, ambient RNA only. 24 Doublet: two cells in one barcode. 25 Ambient RNA: free RNA contaminating droplets. 26 QC metrics: genes, UMIs, MT fraction. 27 Mitochondrial genes: MT- genes; high fraction suggests damage. 28 Ribosomal genes. 29 Normalization. 30 Log1p. 31 HVG: highly variable genes. 32 PCA. 33 kNN graph. 34 Leiden/Louvain: community detection. 35 Resolution. 36 UMAP/t-SNE: 2D visualization. 37 Marker gene. 38 Cell type annotation. 39 Batch effect: technical difference between runs. 40 Donor effect: differences between individuals. 41 Biological replicate: independent donor/animal. 42 Pseudoreplication. 43 Pseudobulk: summed counts per donor per cell type. 44 Differential expression. 45 Log2 fold change. 46 p-value. 47 FDR / adjusted p-value (Benjamini-Hochberg).

48 Mixed-effects model: includes donor random effect. 49 Differential abundance / compositional data. 50 GSEA/gene set enrichment, with TF activity and cell-cell communication as the next-layer interpretations. Extra for T2D biology (not counted): beta cell, alpha cell, delta cell, PP cell, insulin, glucagon, somatostatin, insulin resistance, beta-cell dedifferentiation, ER stress, oxidative stress, islet, HbA1c.

Open items

- Verify GEO metadata for GSE86473, GSE154126; E-MTAB-5061; HPAP access path.
- Find 2022-2026 diabetes single-cell reviews.
- Find a T2D-labelled adipose or other metabolic dataset.
- Confirm pydeseq2/NEBULA/dreamlet specifics from official documentation before using.